A review of instrumentation and measurement in electrodiagnostic medicine is continued in this Part II which focuses on digital instrumentation principles, gain and sweep effects, noise, nerve stimulation, and conduction measurement limitations. With the adoption of microprocessor-based equipment, the neurophysiologic signal must undergo analog-to-digital conversion (ADC) before analysis and display on a video monitor. ADC resolution and sampling rates affect accuracy and measurement precision. Following waveform display, the visual assessment of latency and duration may be influenced by sweep and gain settings, often overlooked sources of error. Undesired signal or noise typically originates from power-line interference, electronic amplifier noise, background muscle activity, or nerve stimulation artifact. Noise often interferes with clinical studies but techniques exist to reduce noise to acceptable levels in virtually all situations. An awareness and understanding of these technical issues will lead to an appreciation of the limitations of electrodiagnostic testing and improve interpretation and clinical decision-making. © 1995 Andrew J. Gitter, MD, and Walter C. Stolov, MD. Published by John Wiley & Sons, Inc.
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AAEM MINIMONOGRAPH #16: INSTRUMENTATION AND MEASUREMENT IN ELECTRODIAGNOSTIC MEDICINE—PART II

ANDREW J. GITTER, MD, and WALTER C. STOLOV, MD

This part of minimonograph #16 continues a discussion of basic electrodiagnostic instrumentation and addresses amplifier basics, waveform display issues, noise, electrical stimulation of nerve, and experimental error in performing nerve conduction velocity (NCV) studies. Two appendices cover patient safety concerns related to nerve stimulation and leakage current.

AMPLIFIER AND DISPLAY CONSIDERATIONS

Electrode–Amplifier Impedance Relationships. A simplified circuit representing the action potential generator (axon or muscle fiber), electrode, and amplifier is shown in Figure 1. The electrode impedance as illustrated includes both the intrinsic impedance of the electrode and the electrode–skin (or electrode–muscle) interface. To measure an action potential, current generated by the depolarizing axon or muscle fiber must flow through the electrode into the amplifier and return to the patient through the ground lead. Because of electrode impedance (resistance) to current flow, there will be a drop in the voltage of the action potential signal across the electrode. As a result, the action potential seen by the amplifier will be attenuated (point B on Fig. 1) relative to its actual amplitude at the electrode tip. To minimize this attenuation, the impedance of the amplifier must be much greater than the electrode impedance. The amount of signal attenuation is determined by the amplifier-to-electrode impedance ratio.

$$\text{Volts}_{\text{amplifier}} = \text{Volts}_{\text{electrode}} \times \left(\frac{\text{Impedance}_{\text{amplifier}}}{\text{Impedance}_{\text{amplifier}} + \text{Impedance}_{\text{electrode}}} \right) \quad (1)$$

To limit the drop in signal amplitude to 1%, the

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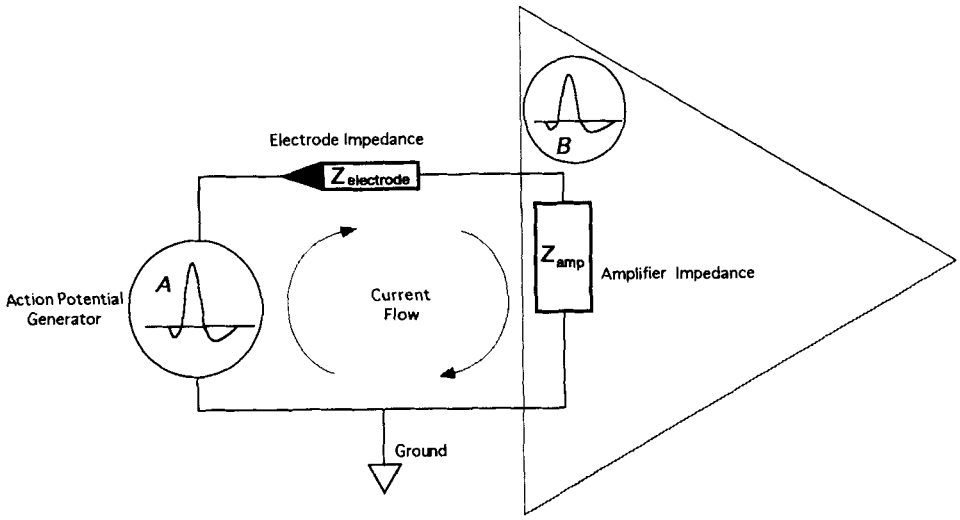


FIGURE 1. Equivalent circuit of a single electrode and the input of an electrophysiologic amplifier. The voltage measured between any two points is equal to the current multiplied by the impedance between the two sites ($V = I \times Z$). Current generated by the depolarization of a nerve or muscle will flow through both the electrode ($Z_{electrode}$) and the input stage of the amplifier ($Z_{amplifier}$). The voltage (V) seen at the amplifier input (site B) is equal to the current (I) flow times the input impedance (Z) of the amplifier ($V = I \times Z_{amplifier}$). The magnitude of this signal is smaller than the actual action potential generated (site A) because of a voltage drop which occurs as the current flows through the recording electrode ($V_{drop} = I \times Z_{electrode}$). By making the amplifier impedance much larger than the electrode impedance this voltage drop can be made negligible.

impedance of the amplifier must be maintained 100 times larger than the impedance of the electrode.

Electrode impedance will vary with frequency and between different types of electrodes. In general, concentric needles will have a higher impedance than monopolar needles which in turn have a higher impedance than surface electrodes. The impedance of the electrode and the amplifier both decrease as frequency increases. To minimize waveform distortion and improve noise rejection, the 100:1 ratio of electrode-to-amplifier impedance should be maintained across the range of frequencies contained in the waveform under study. Impedance as high as 1 MΩ[27] at low frequencies can occur with concentric needles. Thus, to prevent signal degradation, amplifiers should have a minimum input impedance of at least 100–200 MΩ. With modern amplifiers, input impedance is less of a problem. An input impedance of 200–1000 MΩ in parallel with 5–10 pF of capacitance is common. Because of this capacitive component, the input impedance will decrease as signal frequency increases. As an example of this effect, 5 pF of capacitance will decrease the input impedance to a maximum of 320 MΩ at 100 cps, 32 MΩ at 1000 cps, and only 3.2 MΩ at 10,000 cps. Amplifiers designed exclusively for low-impedance surface electrodes used in somatosensory evoked potential (SEP) studies may not be suitable for needle EMG applications.

The ability to reject undesired common mode noise also depends on both electrodes (active and reference) being electrically matched. Unequal electrode impedance unbalances the electrode–amplifier input divider networks, converting some of the common mode noise into a difference signal that is amplified to the same extent as the neurophysiologic signal. Regardless of the technique used, surface, ground, and reference electrodes must not be corroded or dirty and there should be adequate skin preparation to keep impedance low.

Digital Instrumentation Principles. Two techniques of waveform display are commonly employed in electrodiagnostic equipment: analog oscilloscope displays and computer-based digital video display systems. When analog display techniques are used, the neurophysiologic signals can be directly displayed on an oscilloscope screen following amplification and filtering. This method of display results in a continuously varying waveform that is viewed in real time. This approach offers the advantage of a high-quality waveform image (i.e., smooth appearing) but is limited by difficulty in making measurements of various waveform parameters.

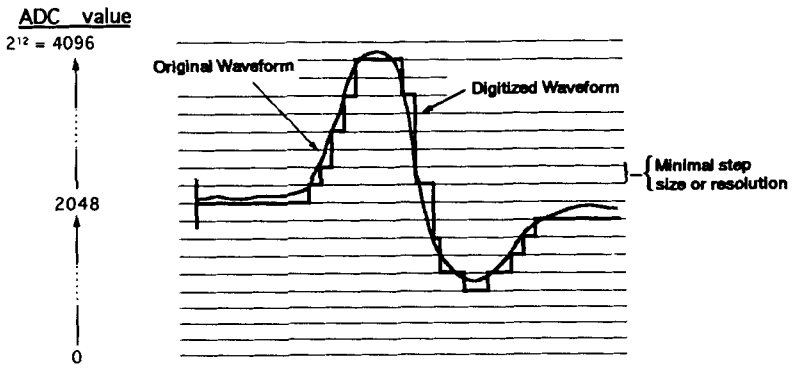
An alternative approach uses analog-to-digital conversion (ADC) and digital signal processing (DSP) techniques. Most contemporary electrodiagnostic equipment uses this method. Following amplification and filtering, the continuously varying

neurophysiologic signal is sampled at discrete time intervals and the amplitude of the signal is converted to a number (digital value). This process is called analog-to-digital conversion. ADC serves as the link or interface between the analog amplifiers and filters and the digital microprocessor-based portions of the instrument. If the sampling rate is fast enough, a series of digital values is generated which provides a close representation of the original analog signal. The advantage of using ADC and DSP is the increased ability to measure various features of the signal after it has been acquired. Changing the sweep and display sensitivity after waveform capture, automatic cursor placement, amplitude and area measurements, averaging, frequency analysis, and interference pattern analysis are examples of the features that are possible with digital waveform manipulation.

The resolution of the measurements made with ADC techniques is determined by the input voltage range and the number of bits of resolution of the analog-to-digital converter. A bit is a single binary digit, i.e., a 0 or 1. Groups of binary digits or bits are used to represent numerical values such as the changes in voltage of a neurophysiologic potential. Groups of binary bits are called bytes (8 bits) or words (16 or 32 bits). Typically, an ADC with 12 bits of resolution is used which can resolve 2^{12} or 4096 discrete amplitude levels. The actual voltage that corresponds to each of these discrete levels will vary depending on the amplifier gain. Effective resolutions will range from less than 1 μV to hundreds of microvolts as the gain changes from its highest to lowest level (Fig. 2).

Following the acquisition of a waveform, the numerical values representing the waveform amplitude over time are stored in electronic memory. In order to actually display the acquired waveform, the data values must be scaled so that an image of the waveform can be represented by a series of closely spaced dots or pixels on a video monitor (Fig. 3). In current electrodiagnostic instruments, the resolution of data acquisition is usually better than video monitor display resolution. The video monitor will typically have a full-scale display of 10 divisions with 30–50 vertical pixels per division. Thus the monitor is only capable of displaying 300–500 steps of amplitude resolution. If a 12-bit ADC was used to digitize the full-scale waveform, the 4096 available steps of digitized resolution must be scaled down to be displayed at the more limited screen resolution. Subtle changes in waveform features and morphology may be lost because of the limited number of pixels used to display the image; however, the advantage of this approach is that a captured waveform can be redisplayed at a greater sensitivity (i.e., 1 mV/div rather than 5 mV/div) without any loss in waveform accuracy. Electrodiagnostic instruments that allow this type of waveform manipulation most commonly use 12–16-bit analog-to-digital converters.

The series of digital numeric values which result from ADC must accurately reflect the ampli-



Amplitude step size for a 12 bit A to D Convertor

Gain s etting	10μV / div	20mV / div
Max. waveform (10 div)	100μV	200mV
ADC step size(resolution)	$\frac{100\ \mu V}{4096\ steps} = 0.0244\mu V$	$\frac{200\ mV}{4096\ steps} = 48.8\mu V$

FIGURE 2. Resolution of the analog-to-digital conversion process. The resolution or stepsize of A–D conversion depends on the full scale amplitude range of the input and the number of bits of the ADC. Since the number of bits is fixed, increasing the gain will allow smaller changes in the waveform to be resolved.

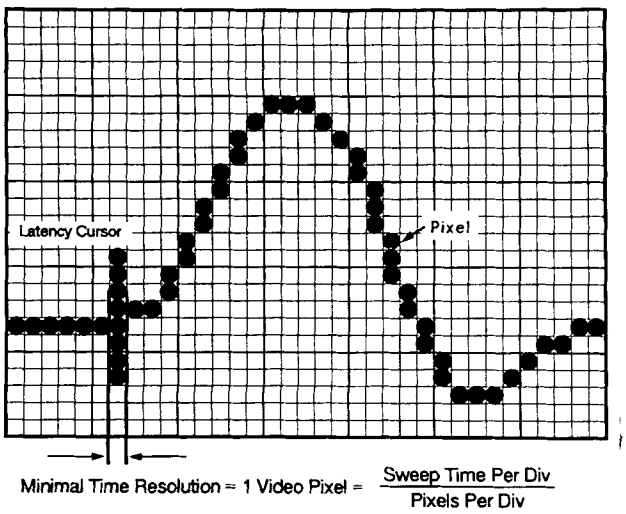


FIGURE 3. Waveform display resolution. Waveforms are displayed as a series of pixels or dots on a video monitor with the visual appearance depending on the number of pixels used to generate the image. The minimal latency or duration interval that can be appreciated is a single horizontal pixel whereas a single vertical pixel is the minimal amplitude change that can be seen.

tude, morphology, and variability in the actual neurophysiologic waveform. This requires that the sampling rate (rate of analog-to-digital conversions) be fast enough to capture the high-frequency, rapidly changing features of the waveform. A minimum sampling rate of more than twice the highest frequency component present in the waveform is needed to preserve all of the waveform information (Nyquist criterion). This sets the lower limit for the analog-to-digital conversion rate. If the sampling rate is insufficient, a form of signal distortion called "aliasing" occurs. This is illustrated in Figure 4. When a signal is sampled at too slow a rate, the signal components which occur at high frequencies will appear in the digitized data folded back at a lower frequency, that is, they appear under an "alias" at a lower frequency.

Simply achieving the Nyquist sampling rate may still not be sufficient to ensure that the morphology of the waveform is accurately reproduced after it is reconstructed on the video monitor as a pattern of pixels or connected points. Although digital data sampled at the Nyquist rate can in theory be used to reconstruct the waveform, the reconstruction process must be appropriately frequency limited. One way of doing this is to transform the series of digital values obtained by ADC to the frequency domain using the Fourier transform and then to reconstruct the waveform with the inverse Fourier transform. This approach is mathematically complex and does not work well for continuous signals such as free-running EMG, so instead, a much simpler method of creating the displayed waveform image is used. A "first order data reconstruction" method is used which consists of drawing a picture of the waveform by connecting the dots that correspond to the individual data samples. This process is conceptually simple and can be done very quickly as demonstrated by the rapid continuous waveform display of contemporary EMG machines. The trade-off is that the signal must be sampled at a more rapid rate than is dictated by the Nyquist criterion. To reconstruct accurately the high-frequency components of the waveform using this technique (connect-the-dots) requires a sampling rate of 5–10 times the highest frequency component.

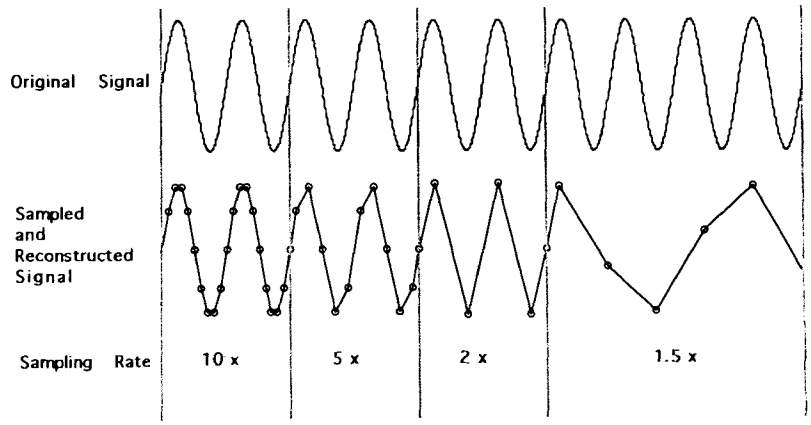


FIGURE 4. The effect of sampling rate. Sampling rates of 5–10 times the signal frequency result in visual reproductions that closely resemble the actual signal. When sampling at the Nyquist rate (twice the signal frequency), frequency information is preserved, but when the waveform is reconstructed by linear interpolation (i.e., drawing a straight line between the individual samples) significant distortion occurs, making the signal look like a triangle wave. Sampling rates less than twice the signal frequency can result in severe distortion and unrecognizable waveforms.

Gain and Sweep Settings Effect. Both the horizontal (sweep) and vertical (sensitivity) setting can influence latency and duration measurements of action potential waveforms.[5,8,14]

Visual appreciation of the point of departure of an action potential from the baseline is influenced by display sensitivity. High sensitivity or gain settings permits visualization of small deflections that are not seen at lower sensitivities. As demonstrated in Figure 5, an increase in the display sensitivity will allow smaller takeoffs from the baseline to be detected, resulting in a shorter onset latency.[8,14] Latencies obtained with a display sensitivity of 5 mV/cm may be as much as 0.5 ms longer than the same potentials measured at a sensitivity of 1 mV/cm.[14] Changes in peak latency are less affected by gain changes because the peak of the waveform is generally more easily identified over a range of settings.

A similar effect of display sensitivity occurs during the measurement of motor unit potential durations and is potentially a major source of error. As the waveform image is made larger by increasing the display sensitivity, the measured duration of the potential will increase as smaller and smaller deflections from the baseline are visualized (Fig. 6). This further complicates the quantitative measurement of motor unit action potential (MUAP) parameters and highlights the need to use standardized procedures for these measurements.

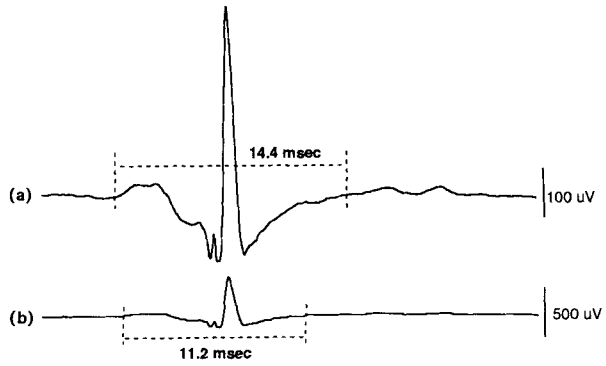


FIGURE 6. The effect of display sensitivity on MUAP duration. With an increase in the display sensitivity, smaller deviations from the baseline are apparent, resulting in a longer visually measured duration.

Changing the sweep setting has a less consistent effect on onset and peak latencies. Although typically, the onset latency will decrease as the sweep speed is increased[8] (i.e., changing the sweep from 5 ms/div to 2 ms/div), this effect is variable and generally of small magnitude.

When using digital EMG instruments, the horizontal display resolution will be limited by the number of horizontal pixels in each grid division. Typically, the video monitor will be capable of displaying 50–100 horizontal pixels per division. The waveform undergoes ADC at a rate that corresponds to the maximal horizontal pixel display requirements. As an example, if the fastest allowed

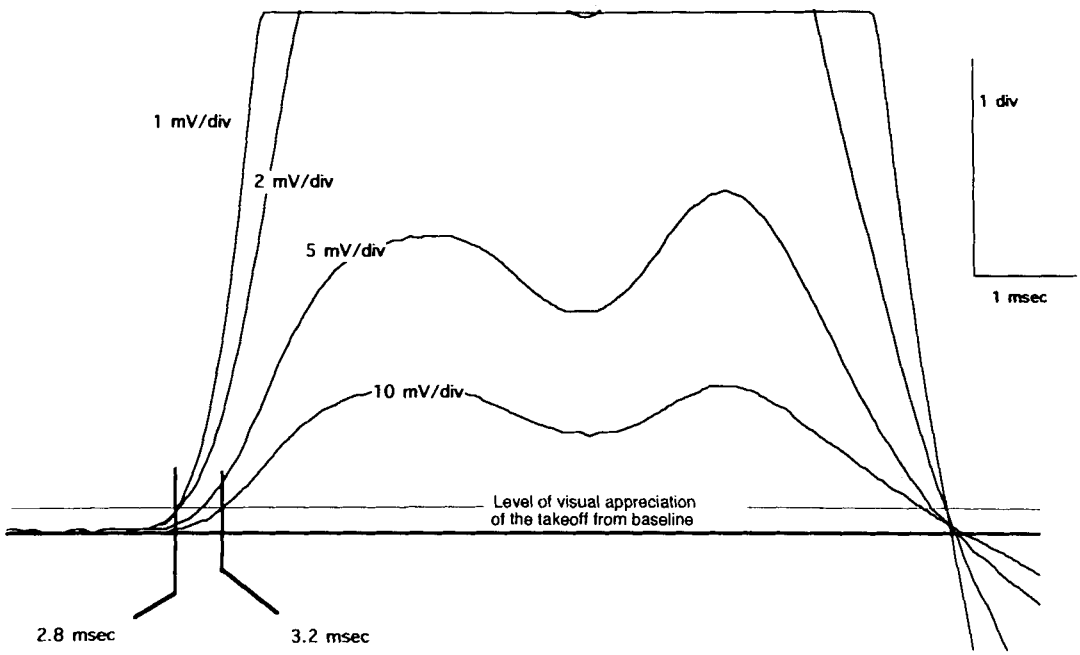


FIGURE 5. Effect of the display gain on the onset latency of a normal median CMAP. As display sensitivity is increased, a smaller deflection from the baseline of the waveform can be observed shortening the visual determined onset latency. In this example, a change in sensitivity from 10 mV/div to 1 mV/div shortens latency by 0.4 ms.

sweep is 1 ms/div and 50 horizontal pixels are displayed, a sampling rate of 50 samples per millisecond or 50,000 cps is needed to acquire 1 data sample for every horizontal screen pixel. When the sweep setting is changed to a slower sweep (i.e., from 1 ms/div to 2 ms/div), just 25 samples are needed for each 1-ms portion of the waveform. To redisplay the waveform at the 2 ms/div setting, only every other digitized waveform value needs to be displayed.

To make latency and duration measurements, a cursor is superimposed onto the waveform image displayed on the video monitor. The smallest cursor adjustment that can be visualized is movement of a single pixel forward or backward. Thus the minimal time interval or change that can be measured is the time associated with a single horizontal pixel. As before, if the display contains 50 pixels per division and the sweep is set at 2 ms/div, each pixel corresponds to 0.04 ms (2 ms/50 pixels = 0.04 ms). When making a measurement, the resolution of the measuring device should be at least twice that of the smallest change that is important to detect. Thus, to report a latency value with a clinical accuracy of 0.1 ms, a cursor resolution of 0.05 ms or less is needed. For most instruments this criterion can be met using a sweep setting of 2 ms/div or less. At slower sweeps (i.e., 5–10 ms/div), less latency accuracy is to be expected.

Many electrodiagnostic instruments are capable of making automated measurements of waveform latency, amplitude, and duration. When the algorithms that make these measurements use internal unscaled waveform data, which typically has better time and amplitude resolution than the displayed waveform image, the automatic measurements of latency and amplitude are relatively insensitive to the display settings. Excessive noise, large and prolonged stimulus potentials, or physiologic artifacts [e.g., contamination of antidromic sensory nerve action potential (SNAP) recording by motor unit potentials] can confuse the automated measured software, resulting in incorrect placement of markers. Visual verification of marker placement is always needed.

Triggers and Delay Lines. Triggers and delay lines are powerful tools used to isolate and display individual MUAPs. Using these techniques, changes in amplitude, duration, phasicity, and stability can be detected much more reliably than can be done from free-running EMG displays. Delay lines work by continuously sampling and storing into memory the ongoing EMG activity. When an MUAP occurs and exceeds a trigger value, typically set by the electromyographer to detect a peak in the potential, the MUAP is extracted from memory and displayed. The MUAP is displayed so that the trigger event is fixed or "time-locked" relative to the start of the sweep. The stable position of the potential makes averaging and qualitative and quantitative assessments of amplitude, duration, and phasicity accurate and reproducible. Delay lines also are useful in detecting motor unit potentials with late or early components (satellite potentials), instability, and blocking. Changes in waveform shape or blocking of individual spikes may be seen in a variety of neuropathic and neuromuscular junction disorders.[28]

More advanced signal processing techniques have been developed to automatically extract multiple motor units from a low-to-moderate force recruitment pattern.[7] A variety of MUAP identification algorithms are used that rely on both time and frequency domain features of the potential. This trend in quantitative motor unit analysis will become more commonplace as clinical electrodiagnostic equipment offers these features and as their clinical utility is documented. At the current time, there has not been adequate comparison studies of the various techniques to support one technique over another.

Signal Averaging. In certain clinical situations, such as the recording of SNAPs, particularly in patients with neuropathies or during the analysis of somatosensory evoked potentials (SEPs), the signal amplitude may be very small compared to background noise. Waveform averaging can be used in these situations to extract the neurophysiologic signal from the larger noise signal.

To average a neurophysiologic event, it must occur at a fixed time relative to a stimulus pulse or another trigger event which can be detected by the instrument. Following the stimulus pulse or trigger event, the waveform is digitized and stored in memory in the instrument. Sequential responses are mathematically summed, averaged, and subsequently displayed. When the undesired signal (noise) is randomly occurring, signal averaging will attenuate the noise, effectively increasing the size of the signal relative to the background noise. This signal-to-noise ratio (SNR) is related to the number of individual responses averaged:

$$\text{SNR} = \frac{\text{Signal Amplitude} \times \sqrt{\text{Number of Sweeps}}}{\text{Noise Amplitude}} \quad (2)$$

Not all undesired signals are randomly occurring. Episodic noise bursts often occur during SEP studies when the patient moves or tenses muscles in the vicinity of the recording electrodes or when the electrocardiogram signal is picked up by the electrodes. This type of noise can be eliminated by using a sweep rejection criterion. Rejection of a sweep occurs if the incoming signal exceeds a predetermined amplitude usually expressed as a percentage of the input range of the amplifier. Typically, sweeps should be rejected when the signal exceeds 95% of the input range.

Motor and motion artifacts can obscure antidromic SNAP or SEP recordings. These artifacts occur at a consistent time following the stimulus and are not removed by averaging. Sixty-cycle interference can usually be considered as a random signal but can become time-locked when it is an exact multiple of the stimulus frequency (i.e., 2.0, 3.0, 4.0, 5.0 cps). The use of a stimulus frequency such as 2.92 or 4.11 cps or random rate stimuli can prevent this effect.

The dependency of the SNR on the square root of the number of sweeps averaged means that the number of sweeps must increase by a factor of 4 in order to double the SNR. During an evoked potential study, a signal that cannot be identified with 500 stimuli because of background noise will require an additional 1500 stimuli to double the SNR. Time and patient tolerance are practical limits on the number of stimuli that can be used during clinical studies. Therefore, to obtain quality recordings, the use of good technique to maximize the signal amplitude (i.e., skin preparation to lower and match electrode impedance and accurate electrode placement) and to minimize noise (i.e., use of correct filter settings and adequate patient relaxation) is essential.

NOISE AND UNDESIRED SIGNALS

Noise can be considered as any signal that interferes with the recording of the desired waveform. During clinical studies, there are three major sources of noise: (1) electrical interference from power-line sources; (2) electronic amplifier noise; and (3) unwanted background electrophysiologic activity.

Power-line interference (50/60-cps sine wave noise) is frequently the most troublesome source of noise encountered in routine clinical studies. This noise is introduced into the recording system by two mechanisms: capacitive and electromagnetic coupling. Small capacitances in the 3–300 pF range exist between the patient, recording leads, power lines, and earth ground.[16] This capacitive coupling allows small interference currents to flow into the patient, the amplifier circuits, and/or the recording leads. Power-line noise can also be magnetically coupled into the amplifier if a loop is formed by the orientation of the recording electrode wire leads or by a combination of the input leads and the patient. Fortunately, most of the 60-cps interference appears as a common mode signal and is rejected. Some of this interference current will actually flow through the soft tissues between the active and reference electrodes, however, generating a difference voltage that will be amplified and displayed. This component of the power-line interference cannot be eliminated by increasing the common mode rejection ratio (CMRR) or other amplifier characteristics and must be reduced by decreasing the amount of noise that is coupled into the patient. Any active-reference electrode impedance mismatch will also contribute to the 50/60-cps interference signal.

The high-input impedance and CMRR of contemporary electrophysiologic amplifiers eliminate the majority of power-line noise. Additionally there are a number of techniques which can be used by the electrodiagnostic medical consultant to further reduce power-line interference. The most easily applied of these techniques include: (1) reducing skin–electrode impedance at *both* recording electrodes and the ground by abrasive skin preparation; (2) using short recording leads and/or twisted pair leads to reduce magnetically coupled interference; and (3) turning off unnecessary electrical equipment and empirically moving the patient to different locations in the room to find the "quietest" area since 50/60-cycle interference is not uniform within an examining room and is highest near electrical equipment (often including the EMG instrument itself) and power lines within walls. In addition, changing the physical orientation of the amplifier and recording leads can reduce or better match the coupled 60-cycle noise, minimizing its effect on the recorded signal.

Persistent power-line noise may be helped with the following measures, though some are not applicable in every situation: (1) checking for breaks in the electrode wire, which are usually hidden by the wire insulation, swapping leads or using an ohmmeter to identify a bad lead; (2) matching the impedance of the reference and active electrodes, which is most problematic during needle studies because the active recording electrode usually has a different impedance than the reference electrode (when using monopolar needle electrodes, substitution of a second needle inserted subcutaneously as a reference usually improves impedance

match); and (3) electrically shielding the examination room, although this is costly and rarely if ever required with modern electrodiagnostic equipment.

The second noise source, internal electronic amplifier noise, can also be a problem. This noise typically has a broad frequency spectrum and increases in amplitude at higher frequencies. It appears as baseline fluctuations (fuzziness) and an audible hissing when sensitivities of 10–20 μV/div or greater are used (see Fig. 9 of Part I). When recording small-amplitude sensory responses, particularly in diseased nerves, this noise can overwhelm and obscure the potential. Amplifier noise specifications refer to the magnitude of this internally produced noise with typical values ranging from 3 to 10 μV (peak-to-peak). This source of noise can be reduced by using low-noise amplifiers, lowering the high-frequency filter cutoff during sensory and SEP studies, and averaging to improve the signal-to-noise ratio.

The third type of noise is background electrophysiologic activity. It is important to recognize that the terminology "active" and "reference" are conventions which allow for uniformity in recordings between laboratories. From an amplifier standpoint, both inputs are equally active and can contribute to both the signal and noise which are recorded and displayed. Background activity from incomplete muscle relaxation beneath the reference electrode will be amplified and can contaminate and mask the signal being generated at the site of the active electrode. Patient positioning to ensure good relaxation, locating the reference electrode over a tendon or bony area away from potential muscle contraction, and using the audio to monitor for unwanted activity during sensory NCV and SEP studies can be used to minimize this type of noise.

NERVE STIMULATION

Following a stimulus pulse, electrical currents will be distributed in the volume conductor (extracellular space) around the peripheral nerve.[4] At low intensities most of the current flow is in the superficial soft tissues. As stimulus intensity increases, more of the current will enter the axon at the cathode, flow longitudinally, and exit at the anode. Depolarization of the membrane occurs beneath the cathode while hyperpolarization is occurring at the anode. If the stimulus intensity is large enough and its duration is long enough, threshold is reached at the cathode and an action potential is generated. Standard stimulation techniques require the placement of two small surface electrodes (cathode and anode) on the nerve 2–3 cm apart. The cathode is placed closest to the recording electrode. Traditionally, nerve stimulation has been accomplished using electrical stimulators. More recently, magnetic stimulators have been used to allow stimulation of both peripheral nerves and central nervous system neurons.

Two basic types of electrical stimulators are routinely used, i.e., constant-current and constant-voltage stimulators. Electrical stimulators are typically capable of generating voltages between 0 and 300 V, currents ranging from 0 to 60–100 mA, and stimulus pulse durations of 0.1–1 ms. Constant-voltage stimulators deliver a fixed voltage between the anode and the cathode. Because skin impedance varies from stimulus to stimulus and even during the short interval that the voltage is applied, the resultant current flow may not be constant. Constant-current stimulators deliver a constant current (milliamperes) to the patient for a broad range of stimulating electrode impedances. For this reason, the magnitude of the current strength delivered by a constant-current stimulator is more stable than that delivered by a constant-voltage stimulator and is more usable for stimulus-threshold measurements.

Technical Considerations in the Use of Electrical Stimulators. In vitro, following electrical stimulation, nerve fibers are depolarized in order of their size: large-diameter having lower thresholds than small-diameter axons. This relationship, however, is less apparent in vivo because of variability in anatomy, the position of nerve fibers within the nerve, volume conduction spread of stimulus currents, and possibly changes caused by pathology.[18] During clinical studies, complete depolarization of the nerve is needed to insure that the maximal conduction velocity will be measured. This is typically achieved by using supramaximal stimulation. The stimulus strength is increased until the evoked amplitude is maximized and, then by convention, further increased an additional 10–30%.[4,18]

Caution must be used when employing supramaximal stimulus strengths because the spread of excessive stimulus current can shift the actual site of nerve activation. The presumed site of nerve depolarization is beneath the stimulator cathode but when large currents are needed because of nerve pathology the stimulus site may be displaced by up to 12 mm. Using a current strength twice that needed to achieve a just maximum response can reduce the measured latency by 0.2 ms due to

distal stimulation of the nerve (Fig. 7). The problem of current spread and displacement of the actual site of nerve depolarization is particularly problematic when performing short segment "inching" studies where changes of several tenths of a millisecond in latency may be clinically important. Placement of the stimulator directly over the nerve and the use of just maximal intensities can help maintain accuracy.

Large or excessive stimulus currents can also cause inadvertent activation of neighboring nerves. Spread of excitation to neighboring nerves causes changes in waveform shape, amplitude, and latency that are confusing and easily misinterpreted. Though sometimes obvious, this can be quite subtle. Overflow can usually be detected by carefully observing the waveform shape. As stimulus strength is increased, a plateau in the waveform amplitude will be seen with small increments of stimulus intensity. This is followed by a secondary increase in amplitude or a change in waveform shape as the intensity is further increased and overflow begins. Also the appearance of an initial positive deflection in the evoked response not present at lower stimulus intensities may represent a volume-conducted response originating from distant muscles innervated by the inadvertently depolarized nerve. Regions where nerves are in close anatomic proximity such as the wrist, axilla, Erb's point, and the popliteal fossa are sites where overflow stimulation most frequently occurs.

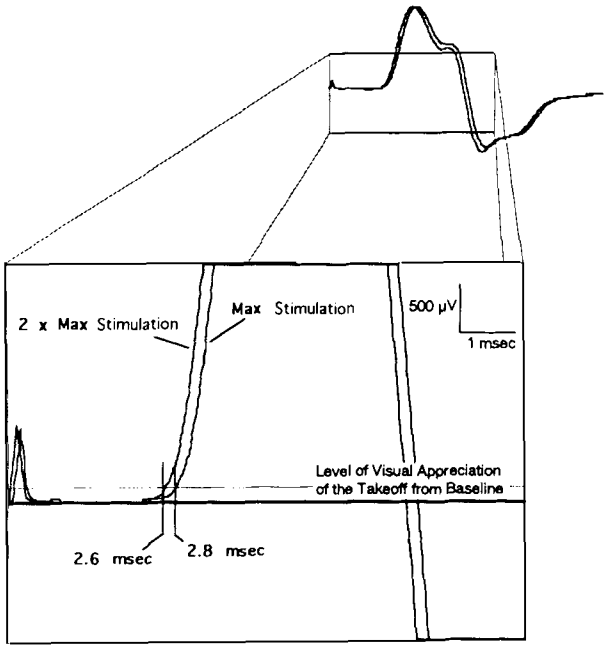


FIGURE 7. Effect of excessive stimulus intensity on the onset latency of a normal ulnar CMAP. Changing the stimulus intensity from just supramaximal to twice this value shortens the onset latency by 0.2 ms. Further increases in intensity will cause further reductions in onset latency.

An unintentional reversal of polarity of the stimulating electrodes (i.e., the anode placed closest to the recording electrode) will effect latency and conduction velocity values.[3,8] For example, assuming a 2-cm distance between the cathode and anode and a 50-m/s conduction velocity, an unsuspected stimulus polarity reversal will result in a latency 0.4 ms too long ($latency_{change}$ = anode–cathode distance/NCV). For a 20-cm segment of nerve, an error of this magnitude will result in a 5.5-m/s velocity error, if the polarity reversal occurred only at one of the stimulation sites.

During routine motor and sensory NCV studies, short duration stimulus pulses are preferred to decrease stimulus artifact, minimize discomfort, and reduce the uncertainty in the actual time of nerve depolarization. H-reflex studies are the exception as they are best elicited with stimulus durations between 0.5 and 1 ms.[13,20] The large diameter group Ia sensory afferents within the tibial nerve appear to be preferentially stimulated because of increased sensitivity to longer-duration stimulus pulses.[3,20,25] Using a through-the-knee stimulation technique, H-reflexes can be reliably obtained at relatively low stimulus intensities even in large individuals. By placing the cathode over the tibial nerve in the popliteal fossa and the anode on the medial side of the patella, current flows through the limb, easily depolarizing the tibial nerve.

Stimulus Artifacts. The stimulus artifact is a (large-amplitude) voltage spike with an exponentially decaying tail that is generated as a result of stimulus current flow in the volume conductor near the recording electrodes. When recording small amplitude responses such as the SNAP, the appearance of the SNAP "riding" on the stimulus artifact tail often makes determination of amplitude difficult and may alter latency values. It is generally hard to predict the actual shape, polarity, or amplitude of the stimulus artifact because multiple factors contribute to its generation.[10,15] The first, and an important, mechanism is the flow of stimulus current between the active and reference recording electrodes. This generates a voltage difference between the active and reference input that will be amplified and displayed. Moving the stimulator closer to the recording electrodes increases the magnitude of the stimulus current flow near the recording electrodes, resulting in a more

pronounced artifact. Second, any recording electrode impedance imbalance will convert common mode stimulus signal into a stimulus artifact proportional to the impedance mismatch. Third, capacitive coupling between the stimulating and recording leads contributes to artifact. The artifact due to these three mechanisms tends to be brief in duration (less than 1 ms) and is the initial spike seen at the start of the sweep. The more prolonged, slowly decaying stimulus tail is usually the result of amplifier saturation and slow recovery caused by the storage of charge in capacitive components of the amplifier. A number of techniques that can be used clinically to decrease the stimulus artifact are listed in Table 1.

Substituting a needle electrode for the stimulator cathode reduces the voltage and current required for nerve depolarization, usually decreases artifact, and improves localization of the actual site of nerve stimulation especially for deep nerve segments.[18] Some degree of caution is required when using needles for stimulation because excessive current densities at the tip of the stimulating needle cause hydrolysis and by inference potential tissue damage (Appendix B). There is, however, little direct in vivo experimental evidence to support any recommendations on the safety of using monopolar EMG needles as stimulating electrodes. Empirically, monopolar needles are used frequently without any apparent adverse effect and specific techniques exist which rely on their use as a stimulating electrode.[6,21] The margin of electrical safety can be improved by stripping the Teflon insulation back 3–4 mm to distribute the current over a larger area. Concentric EMG electrodes should not be used because the small area of the central electrode core leads to much higher current densities.

ERROR SOURCES AND NORMAL VALUES

Normative values for NCV (and MUAP) parameters will vary between studies, in part because of differences in sweep, gain, and filter settings. One setup is not necessarily better than another; standards of normal and abnormal are simply dependent on the particular settings used. Using normal data from other laboratories or from published studies, without adopting similar techniques and corresponding instrument settings, can result in misleading results. Even within a single laboratory, normal values will vary if amplifier sensitivity is altered. Since it is unrealistic for all laboratories to establish their own normal values, individual electromyographers should assure themselves that normal values obtained with their usual NCV and EMG techniques conform to published values before such "normal" values are blindly adopted. Recently, even the use of the mean ± 2 SD values for NCV latency and amplitude measurement has been questioned.[22] Waveform parameters for many commonly performed nerve conduction studies are not normally distributed and use of raw data from control subjects in determining normal limits may result in a high rate of misclassification of patient results.

For conduction velocity determinations there

Table 1. Techniques used to reduce stimulus artifact during NCV studies.

Minimize the current needed for nerve depolarization.	Use short duration and just supramaximal stimuli.
	Decrease stimulator–skin impedance by abrasive cleaning of the stimulation site.[15]
	Substitute a needle electrode for the stimulus cathode.*
Decrease the stimulus current and voltage gradient at the recording electrodes.	Use electrically isolated stimulators (standard on most modern instruments).
	Decrease and match recording electrode impedance with abrasive skin preparation.[15]
	Position the ground between the stimulator and recording leads.
	Remove perspiration and excess electrode paste to prevent stimulus current conduction along the skin surface.[24]
	Rotate the stimulator anode to adjust the orientation of the stimulus equipotential lines until the artifact is minimized.[10,12,24]
	Use short stimulator and recording leads, widely separated from each other, to minimize capacitive coupling.[10]
Stimulator design techniques.	Constant current stimulators generally have smaller artifacts compared to constant voltage stimulators.[10]
	Use amplifier blanking circuits,[1,9,23] direct coupled amplifiers,[26] biphasic stimulus pulses,[12,17] and/or digital stimulus artifact subtraction.[2]

**See text for caution regarding use of needle stimulation.*

are additional sources of error due to inaccuracies in distance (D) and latency (t) measurements.[14] The importance of these errors is under-appreciated and can lead to misinterpretation of clinical studies particularly when "abnormalities" are mild or borderline. The experimental error in a conduction velocity (CV) measurement is given by the expression:

$$\Delta CV = \left(\frac{\Delta D}{D} + \frac{\Delta t}{t}\right) \times CV \qquad (3)$$

where ΔCV = conduction velocity error, ΔD = distance error, and Δt = latency error.

Errors in time (latency) measurements between two stimulation sites result from the factors previously discussed: uncertainty over the actual waveform departure from baseline, gain and sensitivity effects, filter settings, stimulus artifacts, and excessive stimulus intensity. Distance measurements have fewer sources of error and generally contribute less to the CV error than do time errors. Distance measurement errors, however, become especially problematic in several situations: (1) when very short interstimulus distances are used such as occurs in inching studies ($\Delta t/t$ errors also increase in this situation); (2) when the anatomic course of the nerve is uncertain as occurs following nerve transpositions; and (3) when the nerve segment under study crosses the extensor surface of a joint (i.e., ulnar nerve at the elbow). In this situation redundancy in the nerve makes measurements of distance dependent on the joint angle.

Errors of 0.5 ms in time difference and 3–4 mm in the distance are reasonable estimates of measurement error.[14] As can be inferred from eq. (3), the clinical significance of these errors depends on the magnitude of the latency and distance error *relative* to the total segment latency change or distance. For example, a 4-mm error over a 2-cm distance (as used in inching studies) represents a 20% error, whereas the same error over a 20-cm nerve segment is only a 2% error. Figure 8 illustrates the effect these errors will have on CV calculations. These error considerations also imply that a single latency measurement at a fixed distance would yield a more accurate measure of disease progress over time than would a conduction velocity determination.

Not discussed in this monograph are additional sources of variability in NCV and EMG studies including temperature and age effects.[11,19] The many sources of potential error and variability when acquiring and measuring neurophysiologic

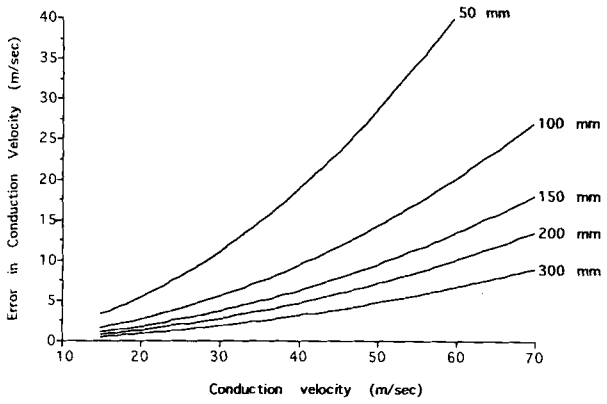


FIGURE 8. Error in conduction velocity resulting from inaccuracies in the measurement of interstimulus latency difference and distance. The family of curves reflect the CV error for a conduction time error of 0.5 ms and a distance error of 3.6 mm for interstimulus distances ranging from 50 mm to 300 mm.

potentials dictates that a reasonable degree of caution be used in evaluating mild or borderline changes, especially when mild abnormalities occur in neurologically normal patients or an abnormality exists in isolation (i.e., mild slowing without other signs of demyelination such as temporal dispersion or conduction block).

SUMMARY

This monograph has explored the influence of instrumentation and measurement issues on the detection, acquisition, display, and characterization of nerve and muscle action potentials encountered during clinical studies. The major areas covered included:

1. Distortions in waveforms that result from inadequate high- and low-frequency response.
2. The influence of electrode characteristics on waveform parameters.
3. Effect of amplifier gain, sensitivity, and sweep on waveform measurements.
4. Noise sources and techniques used to minimize this source of error.
5. Basics of nerve stimulation.
6. Technical and measurement factors affecting the accuracy of conduction velocity studies.

An appreciation of the sources and magnitudes of the various errors and techniques which can be used to minimize their effects will improve the electrodiagnostic physician's ability to acquire accuracy data. Knowledge of these technical limita-

tions will prevent overinterpretation of subtle and borderline abnormalities that may merely represent artifact or "experimental" error associated with performing a clinical study of a nerve or muscle.

APPENDIX A

Leakage Current and Patient Safety. An electrodiagnostic instrument can be a potential source of electrical shocks to both patients and equipment users. Electrical equipment is typically powered by 110-V, 60-cps (220-V, 50-cps in Europe) line current. The electrical connections at a wall outlet consist of a "hot" wire at 110 V, a neutral wire connected to earth ground, and a third safety ground also connected to earth ground. Within the EMG machine a power supply converts this line voltage to lower DC voltages, typically between 5 and 15 V, which are used to power the amplifiers, filters, and computer circuits. Direct physical contact with the high voltage points in the power supply sources is extremely unlikely because these circuits are physically isolated from any part of the equipment that the user or patient contacts. Because insulators are not perfect and alternating currents can be capacitively coupled from the power-line cords or power-supply transformers, however, small "leakage currents" can flow to the machine chassis and low voltage portions of the instrument.

This leakage current is potentially dangerous. Should a grounded patient touch the chassis, the leakage current will flow through the patient to ground, possibly resulting in harm. This potentially dangerous situation is prevented by using the safety ground which is electrically connected to the instrument chassis and serves as a low-resistance pathway to ground for leakage currents. There are several not uncommon situations where leakage current dangers can still exist. The first of these occurs when the chassis ground wire is broken or disconnected. Now the patient becomes the only pathway to ground for the leakage current and a shock hazard exists. Another potentially dangerous situation can occur when the patient is in contact with other electrical equipment such as an electric bed. Should the safety ground on any of these pieces of equipment fail, leakage current from them can flow through the patient via the ground electrode.

To preserve patient safety, standards have been established for the maximum allowable leakage currents. The ANSI/AAMI standard permits a maximum of 100 μA of leakage current from chassis to ground, and 50 μA of leakage current from patient input lead receptacles to ground (nonisolated inputs). These standards have evolved from experimental studies which show that a 20 μA current at 60 cps, if applied directly to dog heart through a small contact area (such as provided by a cardiac catheter or pacemaker), can induce ventricular fibrillation. All electrodiagnostic equipment should be periodically inspected by a qualified biomedical equipment technician to verify the integrity of safety grounds and to ensure that leakage current limits are not being exceeded.

The preceding discussion is of particular importance to older electrodiagnostic equipment in which the patient ground electrode is actually connected to earth ground. Most contemporary instruments use isolated amplifiers and patient connections. In these machines, all patient leads are electrically isolated from the power lines and earth grounds. This isolation commonly takes the form of optical isolation in which the neurophysiologic signal voltages are converted to an optical signal, transmitted across an electrical barrier, and then reconverted to voltage. Other isolation schemes using capacitive or magnetic coupling are equally effective. The use of isolated connections is a major improvement in patient safety, and makes it extremely unlikely that a dangerous level of leakage current can flow through the patient to ground. Standards for isolated amplifier equipment limit current flow from patient leads to ground to 10 μA, even in the event the patient leads come into direct contact with the hot 110-V 60-cps power line.

APPENDIX B

Hydrolysis Danger during Stimulation with Needle Recording Electrodes. Single randomly selected monopolar and concentric needle electrodes were studied in the author's (WCS) laboratory. The needles were mounted with half the shaft of the needle submerged in 0.9% saline.

In the monopolar experiment the needle was connected to the cathode of a standard constant voltage stimulator. A 1-cm disk electrode immersed in the bath was connected to the anode. Using a constant 1-cps stimulation rate, the voltage and then the pulse duration were increased while measuring current flow. At a voltage strength of 25 V and a pulse duration of 0.2 ms, hydrolysis was visually noted. The peak current was measured at 23 mA. From tip area measurements the peak current density was calculated to be 33 A/cm^2. The energy released with each pulse was calculated to

be 52 cal/cm^3. Visible sparking (ionization) of the hydrogen appeared at a voltage strength of 270 V with a 0.5-ms pulse duration.

In the concentric electrode experiment, the core was connected to the cathode and the cannula to the anode of the stimulator. Hydrolysis was noted at 160 V for a 0.1-ms duration and at 75 V for a 0.2-ms pulse. Ionization appeared at 280 V for a 0.1-ms pulse, and at 125 V when pulse duration was 0.2 ms.

The stimulus intensity at which hydrolysis or ionization occurs for a given needle will vary with its actual tip surface area. The greater the area, the greater the voltage and pulse duration required. While it is not known what tissue damage may result from hydrolysis and/or ionization, monopolar EMG needles can be used with a greater margin of safety if the insulation is stripped 3–4 mm back from the tip. Concentric EMG electrodes cannot be so modified and hence should not be used for stimulation.

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